

Linear Regression from Scratch — Full Worked Example

Linear Regression, Step by Step: From Raw Data to a Prediction

Dataset used throughout: $(1, 2), (2, 4), (3, 5), (4, 7)$

Step 1 — Dataset

We are given four data points, each of the form (x_i, y_i) :

$$(1, 2), \quad (2, 4), \quad (3, 5), \quad (4, 7)$$

Here x is the **independent variable** (input/feature) and y is the **dependent variable** (output/target). The number of observations is $n = 4$.

Terms: data point, observation, sample size n , feature, target.

Step 2 — Plot

Plotting the four points $(1, 2), (2, 4), (3, 5), (4, 7)$ on a graph shows them rising roughly steadily as x increases — an approximately straight-line (linear) trend. This visual check is why we choose a **linear model** rather than, say, a curve.

Terms: scatter plot, trend, linearity assumption.

Step 3 — Vectors

Collect all x values into one column vector, and all y values into another:

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 7 \end{bmatrix}$$

Terms: column vector, feature vector (x), target/response vector (y), transpose (written T , turns a column into a row and vice versa).

Step 4 — Hypothesis

We propose that y can be approximated by a straight-line function of x :

$$\hat{y} = \beta_0 + \beta_1 x$$

- $\hat{y}$ (read “y-hat”) is the **predicted** value, as opposed to the actual observed y .
- β_0 is the **intercept** (value of $\hat{y}$ when $x = 0$).
- β_1 is the **slope** (how much $\hat{y}$ changes per unit increase in x).

Terms: hypothesis function, predicted value $\hat{y}$, intercept β_0 , slope/coefficient β_1 .

Step 5 — Design Matrix

To write the hypothesis for **all four points at once** using matrix multiplication, we build a **design matrix** X : one row per data point, first column all 1’s (to multiply the intercept β_0), second column the actual x values (to multiply the slope β_1):

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

Terms: design matrix, bias/intercept column, feature column.

Step 6 — Parameter Vector

Stack the two unknowns we are solving for into a single vector:

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

Terms: parameter vector, weights, coefficients.

Step 7 — Linear Combination (Matrix Form of the Hypothesis)

Multiplying X by β reproduces the hypothesis for every row simultaneously:

$$\hat{y} = X\beta = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 1 \cdot \beta_0 + 1 \cdot \beta_1 \\ 1 \cdot \beta_0 + 2 \cdot \beta_1 \\ 1 \cdot \beta_0 + 3 \cdot \beta_1 \\ 1 \cdot \beta_0 + 4 \cdot \beta_1 \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 \\ \beta_0 + 2\beta_1 \\ \beta_0 + 3\beta_1 \\ \beta_0 + 4\beta_1 \end{bmatrix}$$

Each row is exactly $\beta_0 + \beta_1 x_i$ — a **linear combination** of the columns of X (each column scaled by a β , then added).

Terms: matrix-vector product, linear combination.

Step 8 — Linear Independence (full derivation)

Terms introduced here: linear combination, trivial solution, linear independence/dependence, invertible matrix, singular matrix, determinant, rank.

The two columns of X are:

$$c_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Definition. c_1 and c_2 are **linearly independent** if the only scalars k_1, k_2 satisfying

$$k_1 c_1 + k_2 c_2 = \mathbf{0}$$

are the **trivial solution** $k_1 = 0, k_2 = 0$. If some *non-zero* pair also works, the columns are **dependent**.

Set up the equation using our actual columns:

$$k_1 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

This expands into four scalar equations, one per row:

$$k_1 + k_2 = 0 \quad (1)$$

$$k_1 + 2k_2 = 0 \quad (2)$$

$$k_1 + 3k_2 = 0 \quad (3)$$

$$k_1 + 4k_2 = 0 \quad (4)$$

Solve using equations (1) and (2). Subtract (1) from (2):

$$\begin{aligned} (k_1 + 2k_2) - (k_1 + k_2) &= 0 - 0 \\ k_2 &= 0 \end{aligned}$$

Substitute $k_2 = 0$ into (1):

$$k_1 + 0 = 0 \implies k_1 = 0$$

Verify against the remaining equations:

- (3): $k_1 + 3k_2 = 0 + 3(0) = 0$ (OK)
- (4): $k_1 + 4k_2 = 0 + 4(0) = 0$ (OK)

Conclusion: the only solution is the trivial one, $k_1 = 0$, $k_2 = 0$. Hence c_1 and c_2 are **linearly independent**.

Why it matters. Linear independence of X 's columns is equivalent to $X^T X$ being **invertible** (non-singular), i.e. $\det(X^T X) \neq 0$. Using the value of $X^T X$ computed in Step 13:

$$X^T X = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}, \quad \det(X^T X) = (4)(30) - (10)(10) = 120 - 100 = 20 \neq 0$$

Since the determinant is non-zero, $X^T X$ is invertible, so the normal equation (Step 12) has a **unique** solution for β . If the two columns had instead been dependent (e.g. if every x_i were the same constant), the determinant would be 0, $X^T X$ would be singular, and the regression would have no unique solution.

Step 9 — Error

For each data point, the **error** (also called **residual**) is the gap between the true value and the model's prediction:

$$e = y - \hat{y}$$

Terms: error, residual, error vector.

Step 10 — Cost Function

To measure how good a given β is across **all** points at once, define the **cost function** (mean squared error, with a conventional $\frac{1}{2n}$ scaling factor that simplifies calculus later):

$$J(\beta) = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad n = 4$$

Terms: cost function, loss function, mean squared error (MSE), objective function.

Step 11 — Minimize the Cost

We want to choose β to make $J(\beta)$ as small as possible. Setting the derivative (gradient) of J with respect to β to zero,

$$\frac{\partial J}{\partial \beta} = 0$$

and solving leads directly to a closed-form equation called the **normal equation** (derived from calculus, not repeated symbolically here since it's a standard result).

Terms: optimization, gradient, first-order condition, closed-form solution.

Step 12 — Normal Equation

The normal equation is:

$$X^T X \beta = X^T y$$

This turns the minimization problem into a straightforward linear algebra problem: solve for β using known matrices $X^T X$ and $X^T y$.

Terms: normal equation, least squares.

Step 13 — Matrix Calculation: $X^T X$

First, transpose X :

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

Now multiply X^T (2×4) by X (4×2) to get a 2×2 result. Each entry is a dot product of a row of X^T with a column of X :

Entry (1,1): row 1 of X^T dot column 1 of X :

$$1(1) + 1(1) + 1(1) + 1(1) = 4$$

Entry (1,2): row 1 of X^T dot column 2 of X :

$$1(1) + 1(2) + 1(3) + 1(4) = 1 + 2 + 3 + 4 = 10$$

Entry (2,1): row 2 of X^T dot column 1 of X (symmetric to (1,2), same value):

$$1(1) + 2(1) + 3(1) + 4(1) = 10$$

Entry (2,2): row 2 of X^T dot column 2 of X :

$$1(1) + 2(2) + 3(3) + 4(4) = 1 + 4 + 9 + 16 = 30$$

$$X^T X = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

(Note $X^T X$ is always symmetric — entry (1,2) equals entry (2,1).)

Step 14 — Matrix Calculation: $X^T y$

Multiply X^T (2×4) by y (4×1) to get a 2×1 result:

Row 1:

$$1(2) + 1(4) + 1(5) + 1(7) = 2 + 4 + 5 + 7 = 18$$

Row 2:

$$1(2) + 2(4) + 3(5) + 4(7) = 2 + 8 + 15 + 28 = 53$$

$$X^T y = \begin{bmatrix} 18 \\ 53 \end{bmatrix}$$

Step 15 — Equations

Substituting into the normal equation $X^T X \beta = X^T y$:

$$\begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 18 \\ 53 \end{bmatrix}$$

This is really two scalar equations:

$$4\beta_0 + 10\beta_1 = 18 \quad (\text{i})$$

$$10\beta_0 + 30\beta_1 = 53 \quad (\text{ii})$$

Terms: system of linear equations.

Step 16 — Gaussian Elimination

From (i): solve for β_0 in terms of β_1 . Divide the whole equation by 4:

$$\beta_0 = \frac{18 - 10\beta_1}{4} = 4.5 - 2.5\beta_1$$

Substitute this expression into (ii):

$$10(4.5 - 2.5\beta_1) + 30\beta_1 = 53$$

Expand the multiplication:

$$45 - 25\beta_1 + 30\beta_1 = 53$$

Combine like terms ($-25\beta_1 + 30\beta_1 = 5\beta_1$):

$$45 + 5\beta_1 = 53$$

Subtract 45 from both sides:

$$5\beta_1 = 8$$

Terms: elimination, substitution, back-substitution.

Step 17 — Slope

Divide both sides by 5:

$$\beta_1 = \frac{8}{5} = \boxed{1.6}$$

Step 18 — Intercept

Substitute $\beta_1 = 1.6$ back into $\beta_0 = 4.5 - 2.5\beta_1$:

$$\beta_0 = 4.5 - 2.5(1.6) = 4.5 - 4.0 = \boxed{0.5}$$

Verification — plug both values into equation (ii) to confirm consistency:

$$10(0.5) + 30(1.6) = 5 + 48 = 53 \text{ (OK)}$$

Step 19 — Regression Line

Putting the intercept and slope into the hypothesis from Step 4:

$$\boxed{\hat{y} = 0.5 + 1.6x}$$

This is the fitted **line of best fit**.

Step 20 — Training Predictions

Plug each original x value back into the fitted line:

x	Calculation	$\hat{y}$
1	$0.5 + 1.6(1) = 0.5 + 1.6$	2.1
2	$0.5 + 1.6(2) = 0.5 + 3.2$	3.7
3	$0.5 + 1.6(3) = 0.5 + 4.8$	5.3
4	$0.5 + 1.6(4) = 0.5 + 6.4$	6.9

$$\hat{y} = [2.1, 3.7, 5.3, 6.9]$$

Terms: in-sample predictions, fitted values.

Step 21 — Residuals

Compute $e_i = y_i - \hat{y}_i$ for each point:

x	y (actual)	$\hat{y}$ (predicted)	$e = y - \hat{y}$
1	2	2.1	$2 - 2.1 = -0.1$
2	4	3.7	$4 - 3.7 = +0.3$
3	5	5.3	$5 - 5.3 = -0.3$
4	7	6.9	$7 - 6.9 = +0.1$

$$e = [-0.1, +0.3, -0.3, +0.1]$$

Note the residuals sum to zero: $-0.1 + 0.3 - 0.3 + 0.1 = 0$. This is not a coincidence — it always holds for least-squares fits that include an intercept term.

Step 22 — Cost

Using the cost formula from Step 10 with $n = 4$:

$$J = \frac{1}{2n} \sum e_i^2 = \frac{1}{2(4)} \left[(-0.1)^2 + (0.3)^2 + (-0.3)^2 + (0.1)^2 \right]$$

Compute each squared term:

$$(-0.1)^2 = 0.01, \quad (0.3)^2 = 0.09, \quad (-0.3)^2 = 0.09, \quad (0.1)^2 = 0.01$$

Sum them:

$$0.01 + 0.09 + 0.09 + 0.01 = 0.20$$

Divide by $2n = 8$:

$$J = \frac{0.20}{8} = \boxed{0.025}$$

Step 23 — New Input

We now want to predict y for an x value **not in the training data**: $x = 5$. Since 5 lies beyond the observed range of 1 to 4, this is technically **extrapolation** (as opposed to interpolation, which stays within the observed range) — extrapolated predictions are generally less reliable than in-sample ones.

Step 24 — Final Prediction

Plug $x = 5$ into the fitted regression line from Step 19:

$$\hat{y} = 0.5 + 1.6(5)$$

Compute the multiplication:

$$1.6 \times 5 = 8.0$$

Add the intercept:

$$\hat{y} = 0.5 + 8.0 = \boxed{8.5}$$

Final answer: for $x = 5$, the model predicts $\hat{y} = 8.5$.

Summary Table

Order	Concept	Result
1	Dataset	$(1, 2), (2, 4), (3, 5), (4, 7)$
8	Linear independence	Columns independent; only trivial solution $k_1 = k_2 = 0$
13	$X^T X$	$\begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$
14	$X^T y$	$\begin{bmatrix} 18 \\ 53 \end{bmatrix}$
17	Slope	$\beta_1 = 1.6$
18	Intercept	$\beta_0 = 0.5$
19	Regression line	$\hat{y} = 0.5 + 1.6x$
22	Cost	$J = 0.025$
24	Final prediction ($x = 5$)	$\hat{y} = 8.5$